

The empirical recurrence for OEIS A304602, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A304602 records “Empirical recurrence of order 92” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 233$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 100$. Its last coefficient is $c_{92} = -2 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304602 is “Number of $n \times 6$ 0..1 arrays with every element unequal to 0, 1, 2, 5 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

32, 45, 131, 515, 1726, 6007, 19634, 66861, ...

The entry states:

Empirical recurrence of order 92 (see link above)

As of the “Last modified” line on the live entry (May 15 2018 13:30:30, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 233$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 233$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 466$ exact terms from $n = 8$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	3	c_{24}	−245222	c_{47}	−5718429	c_{70}	8511
c_2	8	c_{25}	−327196	c_{48}	−1425799	c_{71}	14227
c_3	−26	c_{26}	722330	c_{49}	8478733	c_{72}	−18416
c_4	17	c_{27}	383279	c_{50}	−7286583	c_{73}	−1818
c_5	57	c_{28}	−1470463	c_{51}	2017554	c_{74}	2320
c_6	−328	c_{29}	886246	c_{52}	1662270	c_{75}	−5238
c_7	120	c_{30}	753423	c_{53}	−4378807	c_{76}	1443
c_8	937	c_{31}	−3603394	c_{54}	2509575	c_{77}	−1099
c_9	−1311	c_{32}	2146607	c_{55}	162402	c_{78}	−473
c_{10}	764	c_{33}	2188138	c_{56}	−939917	c_{79}	1121
c_{11}	3488	c_{34}	−5492695	c_{57}	1476531	c_{80}	752
c_{12}	−11477	c_{35}	5805319	c_{58}	−724085	c_{81}	161
c_{13}	1663	c_{36}	−400597	c_{59}	−342514	c_{82}	66
c_{14}	29907	c_{37}	−5648470	c_{60}	355531	c_{83}	−350
c_{15}	−21239	c_{38}	8258291	c_{61}	−303192	c_{84}	−119
c_{16}	−351	c_{39}	−5504034	c_{62}	221144	c_{85}	6
c_{17}	39579	c_{40}	−742958	c_{63}	194688	c_{86}	15
c_{18}	−120966	c_{41}	8131766	c_{64}	−146000	c_{87}	68
c_{19}	−54092	c_{42}	−11492518	c_{65}	6217	c_{88}	24
c_{20}	188646	c_{43}	7487968	c_{66}	−47351	c_{89}	−1
c_{21}	−93279	c_{44}	−189788	c_{67}	−70536	c_{90}	−2
c_{22}	−160911	c_{45}	−9235426	c_{68}	66627	c_{91}	−4
c_{23}	544934	c_{46}	11438902	c_{69}	8275	c_{92}	−2

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 100$, and it is the recurrence OEIS A304602 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1x^{92-1} - \dots - c_{92}$. Its constant term is $-c_{92} = 2$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 23 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -2 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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